

RES-MOD — Mathematical Model and Coupling

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1 Domain Purpose and Scope

2 Domain Deliverables

3 RES-MOD-SYS — System Boundary and Mathematical Model Architecture

3.1 Purpose and Scope

This Deliverable defines the continuous system boundary for the regional tsunami model and the local three-dimensional tsunami–barrier interaction model. It assigns physical responsibility to each model without prescribing software classes, file formats or implementation APIs.

3.2 Research Questions

The local-model integration answers:

1. Which physical region is owned by the local three-dimensional model rather than the regional depth-averaged model?
2. Which boundaries define the local air–water–barrier domain?
3. Which outputs justify activating the local model?
4. Which coupling modes must remain available between the regional and local models?

3.3 Terminology and Definitions

3.4 Literature Evidence

Qin et al. [1] showed that a three-dimensional RANS model is needed when local vertical velocity structure, nonhydrostatic pressure and direct fluid-force extraction are project outputs. Source role: ‘3D-RANSVOF’. Project conclusion: the local barrier domain cannot be represented solely by the regional depth-averaged equations. Limitation: the result supports model separation and output requirements, not the final numerical implementation.

Takase et al. [2] and Romano et al. [3] support separating regional and high-fidelity local domains while exchanging state information across a controlled interface. Source role: ‘3D-COUPLING’. Project conclusion: the architecture shall retain separable regional and local model boundaries with configurable interface placement. Limitation: neither paper directly fixes the finite-volume transfer operators used by this project.

3.5 Governing Relations and Quantitative Definitions

3.6 System Boundary

The regional model boundary is the horizontal two-dimensional depth-averaged tsunami domain. It owns regional propagation, bathymetric transformation, run-up and inundation at the selected case-study scale. The local three-dimensional model owns structure-scale pressure, overtopping, local free-surface topology and FSI where depth-averaged assumptions are insufficient.

The local hydrodynamic domain is:

$$\Omega_F = \Omega_L \setminus \Omega_B \quad (3.1)$$

where Ω_L is the selected local coastal volume and Ω_B is the fixed candidate barrier or defence geometry. The baseline local model is activated for run-up, overtopping, breaking, local separation, pressure, shear traction, force, moment and comparative barrier-performance analysis.

The local boundary is partitioned as:

$$\partial\Omega_F = \Gamma_{in} \cup \Gamma_{out} \cup \Gamma_{side} \cup \Gamma_{top} \cup \Gamma_T \cup \Gamma_B \quad (3.2)$$

Here Γ_{in} is the regional-to-local incident-wave boundary, Γ_{out} is the downstream open boundary, Γ_{side} represents artificial lateral ocean cuts, Γ_{top} is the atmospheric boundary, Γ_T is fixed terrain and Γ_B is the fixed barrier surface. The side boundaries represent unmodelled lateral ocean, not physical walls, unless a later site-specific defence geometry explicitly requires a wall boundary.

3.7 State Variables and Parameters

3.8 Continuous Governing Model

3.9 Source Terms and Closures

3.10 Initial, Boundary and Interface Conditions

3.11 Model Coupling Requirements

The regional model shall provide boundary histories and checkpoint data for local 3D/interface modelling. The final 2D–3D coupling operators, FSI feedback rules and local extraction region remain separate workstreams.

The Kamaishi and Sendai corridor polygons delivered by ‘RES-DAT-SCN’ are data/model interface objects. They determine where terrain, observations and regional replay histories are

clipped and sampled, but they do not change the governing equations. A corridor boundary shall therefore be justified by event-source location, target coast, severe impact observations, defence geometry, topobathymetry coverage and numerical relaxation requirements rather than by arbitrary drawing.

The baseline coupling path is one-way replay:

$$\mathcal{M}_{\text{regional}} \rightarrow \mathcal{M}_{\text{local}} \quad (3.3)$$

where a saved regional state and interface history drive the local three-dimensional model without feedback during candidate screening. A simultaneously coupled mode shall also be retained where reflection, blocking or other local feedback materially changes regional or nearshore quantities of interest.

3.12 Sufficient Conditions for Validity

The regional-to-local separation is sufficient when the coupling interface lies outside the region where vertical acceleration, breaking, local impact pressure or barrier-generated recirculation dominates. It shall be checked by moving the interface and comparing the engineering quantities of interest.

3.13 Candidate Approaches

3.14 Critical Comparison

3.15 Assumptions and Applicability

3.16 Limitations and Failure Conditions

3.17 Project-Specific Conclusions

The project uses a two-model hydrodynamic architecture: a regional depth-averaged NLSWE model for event-scale propagation and a local three-dimensional URANS–VOF model for barrier-scale hydrodynamics. The split is based on physical validity and output requirements rather than software convenience.

3.18 Selected Approach and Rationale

Use a regional 2D NLSWE model for the large horizontal domain and defer local three-dimensional/interface/FSI modelling to the local domain. This preserves the distinction between a two-dimensional horizontal solve and cross-sectional visualisations extracted from that solve.

The selected local system role is:

local three-dimensional incompressible two-phase Navier–Stokes/VOF model for fixed-terrain, fixed-barrier tsunami–barrier interaction, driven by regional NLSWE interface histories and used to extract free-surface, overtopping, pressure, shear, force, moment and energy metrics

3.19 Unresolved Decisions

Open system-boundary decisions are the final case-study local-domain extent, the exact interface location, the downstream buffer length, and the activation gate for simultaneous coupling.

3.20 Requirements and Handoffs

‘RES-MOD-IBC’ shall define the continuous boundary-condition roles on each local boundary. ‘RES-NUM-IBC’ shall convert those roles into discrete boundary-face states and flux treatments. ‘RES-VER-FRM’ shall test that the chosen boundary and coupling placement do not contaminate the local quantities of interest.

‘RES-DAT-CAS’ and Software shall provide the dataset manifest, corridor-polygon identifiers and replay-boundary histories consumed by the mathematical interface. The mathematical model shall treat those as inputs, not as proof that validation has been executed.

3.21 Deliverable Completion Record

4 RES-MOD-EQS — Governing Equations and State Variables

4.1 Purpose and Scope

4.2 Research Questions

4.3 Terminology and Definitions

4.4 Literature Evidence

4.4.1 Saito et al. (2014)

Saito et al. [4] compared nonlinear dispersive, linear dispersive, and nonlinear non-dispersive models of the 2011 Tōhoku tsunami while retaining a common source, computational domain, bathymetry, and observational dataset. The study showed that frequency dispersion was required to reproduce short-period offshore wave components generated by a spatially concentrated source, while nonlinear advection and friction became increasingly important during nearshore transformation and coastal reflection.

The paper is retained as the primary evidence that the nonlinear shallow-water equations

are necessary for nearshore propagation but may be insufficient where the earthquake source retains physically important short-wavelength components.

4.4.2 Qin et al. (2018)

Qin et al. [1] compared a depth-averaged nonlinear shallow-water model with a three-dimensional Reynolds-averaged Navier–Stokes model. The depth-averaged formulation reproduced several bulk inundation quantities but could not represent the vertical velocity variation and nonhydrostatic pressure associated with initial impact. The three-dimensional formulation also permitted hydrodynamic force to be obtained directly by integrating the fluid traction over the structural surface rather than through an empirical drag relation.

The paper is retained as the primary evidence that the local barrier domain requires a three-dimensional parent equation system when pressure, traction, and structural loading are project outputs.

4.5 Governing Relations and Quantitative Definitions

4.6 System Boundary

4.7 State Variables and Parameters

4.8 Continuous Governing Model

4.8.1 Consolidated Governing Systems

The current research phase records three governing systems:

1. a regional two-dimensional nonlinear shallow-water equation system;
2. a local three-dimensional nonhydrostatic incompressible immiscible two-phase Navier–Stokes system;
3. a transient structural-dynamics system with rigid, one-way and two-way FSI fidelity levels.

The regional conserved state is:

$$\mathbf{U}_r = [h, q_x, q_y]^T \quad (4.1)$$

$$q_x = hu \quad (4.2)$$

$$q_y = hv \quad (4.3)$$

where h is water depth, u and v are depth-averaged horizontal velocities, q_x and q_y are depth-integrated momenta, b is bed elevation positive upward, and $\eta = h + b$ is free-surface elevation. The primitive regional variables are:

$$\mathbf{W}_r = [h, u, v, \eta, b]^\top \quad (4.4)$$

$$h = \eta - b \quad (4.5)$$

The regional solve is horizontal and depth averaged in (x, y, t) . Vertical sections are diagnostic visualisations extracted from this two-dimensional state; they are not separate solved cross-sectional models.

The selected regional baseline is the nonlinear shallow-water balance law:

$$\frac{\partial \mathbf{U}_r}{\partial t} + \frac{\partial \mathbf{F}_r}{\partial x} + \frac{\partial \mathbf{G}_r}{\partial y} = \mathbf{S}_b + \mathbf{S}_f + \mathbf{S}_{sp} \quad (4.6)$$

with fluxes:

$$\mathbf{F}_r = \left[q_x, \frac{q_x^2}{h} + \frac{1}{2}gh^2, \frac{q_x q_y}{h} \right]^\top \quad (4.7)$$

$$\mathbf{G}_r = \left[q_y, \frac{q_x q_y}{h}, \frac{q_y^2}{h} + \frac{1}{2}gh^2 \right]^\top \quad (4.8)$$

The bed-slope source term is:

$$\mathbf{S}_b = \left[0, -gh \frac{\partial b}{\partial x}, -gh \frac{\partial b}{\partial y} \right]^\top \quad (4.9)$$

The Manning-type bottom-friction term is retained as:

$$\mathbf{S}_f = \left[0, -gn_M^2 \frac{\sqrt{q_x^2 + q_y^2} q_x}{h^{7/3}}, -gn_M^2 \frac{\sqrt{q_x^2 + q_y^2} q_y}{h^{7/3}} \right]^\top \quad (4.10)$$

Open-boundary sponge damping is part of the established regional model where artificial boundary reflections would otherwise contaminate the domain:

$$\mathbf{S}_{sp,i} = -\sigma_i [h_i - h_{\text{ref},i}, q_{x,i}, q_{y,i}]^\top \quad (4.11)$$

$$h_{\text{ref},i} = \max(\eta_{\text{ref}} - b_i, 0) \quad (4.12)$$

Earthquake forcing is represented by dynamic bathymetry:

$$b(x, y, t) = b_0(x, y) + \sum_m \Delta b_m(x, y) R_m(t) \quad (4.13)$$

Because h is the conserved depth variable in Equation (4.1), this formulation does not add a separate earthquake mass source to the depth equation.

The Coriolis source term remains an optional sensitivity term rather than part of the baseline locked regional model:

$$\mathbf{S}_c = [0, f q_y, -f q_x]^\top \quad (4.14)$$

where the sign convention follows the chosen horizontal coordinate orientation.

Wetting and drying are part of the regional model requirements rather than a separate governing-equation family. A dispersive extension remains optional and shall be activated only where regional comparison shows that nonlinear shallow-water propagation does not retain the required phase, amplitude, or spectral content.

The local fluid governing system remains the three-dimensional nonhydrostatic incompressible immiscible two-phase Navier–Stokes system defined in Section 4.8.2. Its required baseline outputs include breaking, run-up, overtopping, separation, recirculation, pressure and shear traction, URANS turbulence variables and optional LES comparison fields. Moving fluid–structure boundaries belong to deferred FSI activation and are not part of the fixed rigid barrier baseline.

The structural governing system is transient structural dynamics. Its fidelity hierarchy is rigid barrier, one-way CFD-to-FEA, and two-way FSI where feedback is material. The structural formulation shall not be permanently reduced to a single degree of freedom.

The reviewed Tōhoku hindcasts do not support the use of one governing model across every spatial and physical regime. A provisional model hierarchy is therefore defined as:

$$\mathcal{M} = \begin{cases} \mathcal{M}_{\text{source}}, & \text{earthquake-driven tsunami generation} \\ \mathcal{M}_{\text{disp}}, & \text{propagation where frequency dispersion is material} \\ \mathcal{M}_{\text{NLSW}}, & \text{nonlinear nearshore propagation and inundation} \\ \mathcal{M}_{\text{3D}}, & \text{local barrier interaction and vertically resolved flow} \end{cases} \quad (4.15)$$

The hierarchy describes separable modelling regimes rather than a requirement to implement four independent solvers. A sufficiently general model may cover more than one regime, provided that its accuracy and computational cost remain acceptable.

4.8.2 Selected Three-Dimensional Fluid Equations

The selected parent model is the three-dimensional, nonhydrostatic, incompressible, immiscible, two-phase Navier–Stokes system in a one-fluid representation.

Mass conservation is:

$$\nabla \cdot \mathbf{u}_f = 0 \quad (4.16)$$

Momentum conservation is:

$$\frac{\partial (\rho_f \mathbf{u}_f)}{\partial t} + \nabla \cdot (\rho_f \mathbf{u}_f \otimes \mathbf{u}_f) = \nabla \cdot \boldsymbol{\sigma}_f + \rho_f \mathbf{g} + \mathbf{f}_\sigma + \mathbf{f}_{\text{ext}} \quad (4.17)$$

The fluid stress tensor is:

$$\boldsymbol{\sigma}_f = -p\mathbf{I} + 2\mu_{\text{eff}}\mathbf{D}(\mathbf{u}_f) \quad (4.18)$$

$$\mathbf{D}(\mathbf{u}_f) = \frac{1}{2} \left[\nabla \mathbf{u}_f + (\nabla \mathbf{u}_f)^\top \right] \quad (4.19)$$

The water phase is transported according to:

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{u}_f) = 0 \quad (4.20)$$

The mixture density and molecular viscosity are:

$$\rho_f = \alpha \rho_w + (1 - \alpha) \rho_a \quad (4.21)$$

$$\mu_f = \alpha \mu_w + (1 - \alpha) \mu_a \quad (4.22)$$

The effective viscosity is expressed generically as:

$$\mu_{\text{eff}} = \mu_f + \mu_t \quad (4.23)$$

where μ_t is supplied by the selected turbulence treatment.

The term $\mathbf{f}_\sigma$ represents surface-tension forcing. Its physical contribution is expected to be small at tsunami and barrier scales where the Weber number is large. It remains in the general equation set so that its omission can be justified through a regime assessment rather than imposed implicitly.

The term $\mathbf{f}_{\text{ext}}$ permits later local extensions, including locally activated porous resistance or other justified body forces. It is zero in the initial impermeable-barrier model.

Decision trace: the equation set in Equations (4.16), (4.17) and (4.20) is selected as the local parent hydrodynamic model. Source role: ‘3D-RANSVOF’. Supporting sources: Qin et al. [1] and Watanabe et al. [5]. Project-specific conclusion: a three-dimensional nonhydrostatic air–water model is required when pressure, shear, force, moment, run-up and overtopping are outputs. Limitation: the continuous equation selection does not validate the selected mesh, VOF advection, pressure correction, turbulence closure or wall treatment. Downstream handoff: ‘RES-NUM-SPEC’ shall define the finite-volume solution procedure and ‘RES-VER-MET’ shall define acceptance metrics for the extracted hydrodynamic quantities.

4.8.3 Required Hydrodynamic Outputs

The local model shall preserve the spatial fluid traction:

$$\mathbf{t}_f = \boldsymbol{\sigma}_f \mathbf{n} \quad (4.24)$$

The resultant force and moment acting on a structural surface Γ_{fs} are:

$$\mathbf{F}_f(t) = \int_{\Gamma_{fs}} \mathbf{t}_f d\Gamma \quad (4.25)$$

$$\mathbf{M}_f(t) = \int_{\Gamma_{fs}} (\mathbf{x} - \mathbf{x}_0) \times \mathbf{t}_f d\Gamma \quad (4.26)$$

$$\mathbf{J}_f = \int_{t_0}^{t_1} \mathbf{F}_f(t) dt \quad (4.27)$$

The model shall output the complete pressure and traction fields rather than only integrated force coefficients. This is required for conservative transfer to a structural mesh and for calculating:

- local peak pressure;
- pressure duration;
- resultant force;
- force impulse;
- overturning and torsional moments;
- centre-of-pressure migration;
- transmitted discharge and momentum;
- overtopping volume;
- reflected and transmitted wave energy;
- downstream velocity and turbulence fields.

4.9 Local Three-Dimensional Governing-Equation Selection

The regional depth-averaged model is intended to reproduce tsunami generation, propagation, nearshore transformation, and the incident state supplied to the local barrier domain. It is not intended to resolve the complete local interaction between the wave, barrier, free surface, and structural surface.

The evidence retained for this Deliverable establishes two distinct conclusions:

1. a resolved three-dimensional nonhydrostatic model is required for local barrier interaction, pressure, traction, and structural loading;
2. selection of physically complete equations does not remove the requirement to validate the turbulence closure, free-surface treatment, boundary conditions, and numerical implementation.

4.9.1 Candidate Parent Equation Families

The candidate formulations were ranked according to their ability to represent:

- three-dimensional and vertical fluid motion;
- nonhydrostatic pressure;
- wave breaking, run-up, overtopping, and separation;
- spatially and temporally resolved structural traction;
- moving fluid–structure interfaces;
- the current rigid, impermeable, and well-ventilated barrier design space;
- later multiphysics extensions without replacing the parent fluid model.

Published wall-clock times were not included in the equation-family ranking. Computational performance depends on the numerical method, linear and pressure solvers, spatial resolution, hardware, parallel architecture, memory movement, and implementation quality. These quantities will be assessed during numerical-method and software selection rather than inferred from unrelated published simulations.

Reynolds-averaged Navier–Stokes, large-eddy simulation, and hybrid RANS–LES are not treated as separate parent conservation-law systems. They are alternative averaging, filtering, and closure treatments attached to the selected Navier–Stokes equations.

Volume-averaged Reynolds-averaged Navier–Stokes is similarly treated as a porous-region extension rather than as the default parent model.

Particle methods such as smoothed particle hydrodynamics and the moving particle semi-implicit method are numerical formulations. They will be evaluated during numerical-method selection and are not candidates in the present governing-equation ranking.

4.9.2 Selected Fluid State Variables

The operational local URANS state is defined as:

$$\mathcal{Q}_f = \{\mathbf{u}_f, p, \alpha, k, \omega\} \quad (4.28)$$

where:

Rank	Formulation	Principal capability	Project decision
1	Three-dimensional incompressible, immiscible, two-phase Navier–Stokes	Resolves vertical acceleration, nonhydrostatic pressure, free-surface deformation, flow separation, and structural traction	Selected as the parent local fluid equation set
2	Compressible two-phase Navier–Stokes	Adds gas and liquid compressibility, acoustic propagation, and trapped-gas compression	Deferred unless enclosed-air or compressibility effects become material
3	Three-dimensional nonhydrostatic multilayer equations	Introduces vertical structure at lower dimensional complexity than fully resolved Navier–Stokes	Not selected for barrier-scale pressure and FSI owing to limited direct load-validation evidence
4	Three-dimensional hydrostatic multilayer equations	Resolves several vertical layers while retaining hydrostatic pressure	Rejected for local impulsive and nonhydrostatic impact

Table 1. Project-specific ranking of candidate parent fluid equation families for the local barrier domain

- $\mathbf{u}_f$ is the three-dimensional fluid velocity;
- p is the mechanical pressure;
- α is the water volume fraction;
- k is turbulent kinetic energy;
- ω is the specific dissipation rate.

The baseline closure attached to this state is the k – ω SST URANS closure. Source role: ‘3D-SST’. Supporting source: Menter [6]. Project-specific conclusion: the high-Reynolds-number operational local model shall retain explicit turbulence variables rather than treating the simulation as molecular-viscosity-only flow. Limitation: LES and hybrid RANS–LES remain higher-fidelity or later-comparison modes and require their own state variables when activated. Downstream handoff: ‘RES-MOD-SRC’ owns the turbulence closure equations and ‘RES-NUM-SPEC’ owns their discrete solution order.

4.10 Source Terms and Closures

4.11 Initial, Boundary and Interface Conditions

4.12 Model Coupling Requirements

4.13 Sufficient Conditions for Validity

4.14 Candidate Approaches

4.15 Critical Comparison

4.16 Assumptions and Applicability

4.17 Limitations and Failure Conditions

4.18 Project-Specific Conclusions

The current review selects the parent local equation family and defines the regional incident-wave model that will provide the state supplied to that model.

The provisional regional baseline is a nonlinear depth-averaged model with optional dispersive terms. Higher-order far-field corrections shall remain independent modules and shall not be included in the near-Japan benchmark unless observational comparison establishes that they materially affect the required outputs.

4.18.1 Project-Specific Governing-Equation Decision

The initial hydrodynamic architecture shall contain two principal equation families:

$$\mathcal{M}_{\text{hydro}} = \{\mathcal{M}_{\text{regional}}, \mathcal{M}_{\text{local}}\} \quad (4.29)$$

$$\mathcal{M}_{\text{regional}} := \text{selected depth-averaged regional formulation} \quad (4.30)$$

$$\mathcal{M}_{\text{local}} := \text{three-dimensional incompressible two-phase Navier–Stokes formulation} \quad (4.31)$$

The local model shall initially represent a rigid, impermeable, and well-ventilated barrier. Turbulence closure, FSI coupling, porous resistance, and compressible-air treatment are modular attachments whose activation is governed by the physical regime and design geometry.

For the baseline hydrodynamic campaign the closure attachment is no longer optional: the operational local model is URANS with $k-\omega$ SST, while LES is retained for selected high-fidelity comparison cases.

4.19 Model Selection Criteria

The regional governing model shall be selected using:

-
- the spatial and temporal bandwidth of the earthquake source;
 - propagation distance;
 - local water depth;
 - expected wave nonlinearity;
 - the importance of wetting and drying;
 - the required accuracy of arrival time, amplitude, phase, and spectral content;
 - the intended coupling location with the local three-dimensional model;
 - the available computational resources.

The baseline comparison shall evaluate:

$$\mathcal{M}_{\text{NLSW}} \quad \text{and} \quad \mathcal{M}_{\text{NLSW+B}} \quad (4.32)$$

where $\mathcal{M}_{\text{NLSW+B}}$ denotes a nonlinear shallow-water formulation augmented by the selected Boussinesq or equivalent dispersive terms.

The dispersive extension shall be retained only where it produces a material improvement in the benchmark quantities relative to its additional computational cost.

4.20 Selected Approach and Rationale

4.21 Unresolved Decisions

4.22 Requirements and Handoffs

The following conclusions are retained from the comparative studies:

1. nonlinear momentum transport and friction are required in the nearshore region and for waves that have interacted with the coastline;
2. frequency dispersion becomes material when the source contains short-wavelength components or when small phase-speed differences accumulate over sufficient distance;
3. a source model that removes short-wavelength content may conceal the importance of dispersion;
4. time-dependent source generation may alter arrival time and spectral content relative to instantaneous initialisation;
5. seawater-density stratification, elastic loading, and gravitational-potential change principally improve trans-oceanic phase and arrival-time accuracy;
6. no far-field correction shall be enabled solely on the basis that it increases physical complexity.

4.23 Deliverable Completion Record

5 RES-MOD-SRC — Physical Source Terms and Constitutive Closures

5.1 Purpose and Scope

5.2 Research Questions

5.3 Terminology and Definitions

5.4 Literature Evidence

5.4.1 Kozelkov et al. (2022)

Kozelkov et al. [7] compared a two-phase Navier–Stokes calculation without a turbulence closure against a Reynolds-averaged calculation using the k – ω SST model. Differences were small during smooth propagation but became material during breaking, run-up, overtopping, collapse, and barrier loading. The reported changes in total barrier force reached engineering significance in the more energetic cases.

5.4.2 Watanabe et al. (2022)

Watanabe et al. [5] compared blind submissions using laminar, Reynolds-averaged, and large-eddy formulations. Dynamic sub-grid kinetic-energy LES submissions generally produced more consistent force predictions than fixed-coefficient Smagorinsky submissions, while the submitted k – ε calculations were more stable and consistent than the submitted k – ω calculations for that experiment.

5.4.3 Menter (1994)

Menter [6] introduced the k – ω SST eddy-viscosity closure by blending near-wall k – ω behaviour with outer-flow k – ε behaviour and limiting turbulent shear stress. Source role: ‘3D-SST’. Project conclusion: SST is a defensible operational URANS baseline for high-Reynolds-number local tsunami run-up and impact studies. Limitation: this source is a general turbulence-model source; project validation must still test the closure against free-surface, pressure, overtopping and load outputs.

5.5 Turbulence Treatment for the Local Three-Dimensional Model

The selected parent Navier–Stokes equations require a treatment of unresolved turbulent transport in the local impact region.

The local regime is treated as high-Reynolds-number turbulent free-surface impact flow. A molecular-viscosity-only calculation may be used diagnostically but is not accepted as the operational model for candidate screening.

These results do not establish one universally optimal closure. The submitted models differed in their meshes, solvers, interface treatments, wall conditions, and numerical schemes. The findings instead establish that:

1. turbulence closure is required in the local breaking and impact domain;
2. the closure must be assessed within a controlled common numerical framework;
3. free-surface agreement alone is insufficient for selecting the closure;
4. closure selection must include pressure, force, impulse, overtopping, and downstream-flow measures.

5.5.1 Turbulence-Model Hierarchy

The turbulence formulations are assigned the following project roles:

Formulation	Project role	Current decision
Unsteady RANS	Operational local-impact model	Selected as the initial candidate-evaluation formulation with $k-\omega$ SST as the baseline closure
Dynamic sub-grid kinetic-energy LES	High-fidelity comparison model	Retained for validation cases and short-listed geometries
Hybrid RANS–LES	Possible intermediate-fidelity model	Retained for later research; insufficient evidence for current selection
Fixed-coefficient Smagorinsky LES	Alternative LES closure	Retained as a comparator but not provisionally preferred
Molecular-viscosity-only Navier–Stokes	Diagnostic case	Not accepted as the operational high-Reynolds-number impact model

Table 2. Project roles assigned to the candidate turbulence formulations

The operational and high-fidelity formulations shall use the same parent fluid equations, material properties, geometry, incident state, and output definitions wherever practical.

5.6 Governing Relations and Quantitative Definitions

5.6.1 Unsteady Reynolds-Averaged Formulation

The instantaneous fluid velocity is decomposed as:

$$u_i = \bar{u}_i + u'_i \quad (5.1)$$

Averaging introduces the Reynolds stress:

$$\tau_{ij}^R = -\rho_f \overline{u'_i u'_j} \quad (5.2)$$

For a linear eddy-viscosity closure:

$$\tau_{ij}^R = 2\mu_t \bar{S}_{ij} - \frac{2}{3}\rho_f k \delta_{ij} \quad (5.3)$$

$$\bar{S}_{ij} = \frac{1}{2} \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) \quad (5.4)$$

The operational URANS closure is the k - ω SST model. Its generic transport form is recorded as:

$$\frac{\partial (\rho_f k)}{\partial t} + \nabla \cdot (\rho_f \mathbf{u}_f k) = \nabla \cdot [(\mu_f + \sigma_k \mu_t) \nabla k] + P_k - \beta^* \rho_f k \omega \quad (5.5)$$

$$\frac{\partial (\rho_f \omega)}{\partial t} + \nabla \cdot (\rho_f \mathbf{u}_f \omega) = \nabla \cdot [(\mu_f + \sigma_\omega \mu_t) \nabla \omega] + \gamma \frac{\omega}{k} P_k - \beta \rho_f \omega^2 + D_\omega \quad (5.6)$$

The eddy viscosity is limited in SST form by:

$$\mu_t = \rho_f \frac{a_1 k}{\max(a_1 \omega, S F_2)} \quad (5.7)$$

where P_k is the turbulence-production term, S is a strain-rate measure, F_2 is the SST blending function and D_ω contains the cross-diffusion contribution. The coefficient set and blending functions shall follow the selected SST reference rather than being retuned during the research specification.

Decision trace: source role ‘3D-SST’; supporting sources Menter [6], Kozelkov et al. [7] and Watanabe et al. [5]. Project-specific conclusion: URANS k - ω SST is the baseline turbulence closure for repeated local candidate assessment. Limitation: the closure is not accepted for final engineering certification until the validation hierarchy in ‘RES-VER’ confirms pressure, force, impulse and overtopping behaviour. Downstream handoff: ‘RES-NUM-SPEC’ shall define when k and ω are solved in the time-step sequence and ‘RES-VER-MET’ shall define the URANS–LES comparison metrics.

5.6.2 Large-Eddy Formulation

Large-eddy simulation filters the velocity field into resolved and sub-grid components:

$$u_i = \tilde{u}_i + u_i^{\text{sgs}} \quad (5.8)$$

Filtering introduces the sub-grid stress:

$$\tau_{ij}^{\text{sgs}} = \rho_f (\widetilde{u_i u_j} - \widetilde{u_i} \widetilde{u_j}) \quad (5.9)$$

The deviatoric sub-grid stress may be represented as:

$$\tau_{ij}^{\text{sgs}} - \frac{1}{3} \tau_{kk}^{\text{sgs}} \delta_{ij} = -2\mu_{\text{sgs}} \widetilde{S}_{ij} \quad (5.10)$$

A dynamic sub-grid kinetic-energy model is retained as the provisional high-fidelity candidate. Its exact transport equation and coefficient evaluation shall be selected during the dedicated turbulence-closure study.

5.6.3 Turbulence-Closure Selection Gate

The operational closure shall be accepted only after comparison against the high-fidelity formulation for representative impact cases.

The comparison shall include:

- incident and transmitted free-surface histories;
- pressure distributions;
- resultant forces and moments;
- force impulse;
- overtopping volume;
- reflected and transmitted energy;
- downstream recirculation and wake structure;
- resolved and modelled turbulent kinetic energy;
- sensitivity of the barrier-performance ranking.

The operational URANS model may be retained where it preserves the design ranking and the required engineering quantities within the acceptance tolerances defined under ‘RES-VER-MET’.

If the barrier ranking changes materially between URANS and LES, the lower-fidelity model shall not be used as the sole optimisation evaluator until the source of the discrepancy has been resolved.

5.7 System Boundary

5.8 State Variables and Parameters

5.9 Continuous Governing Model

5.10 Source Terms and Closures

The regional two-dimensional NLSWE source catalogue is established at research-specification level. The baseline retained terms are bed slope, Manning bottom friction, dynamic earthquake bathymetry, and optional sponge damping near artificial open boundaries. Coriolis and lateral eddy viscosity are not part of the baseline regional model; they remain optional sensitivity terms if the selected domain scale or validation evidence requires them.

The continuous regional source split is:

$$\mathbf{S} = \mathbf{S}_b + \mathbf{S}_f + \mathbf{S}_{sp} \quad (5.11)$$

where $\mathbf{S}_b$ is the bed-slope source, $\mathbf{S}_f$ is Manning bottom friction, and $\mathbf{S}_{sp}$ is the sponge-layer relaxation source defined in Section 4.8. The earthquake contribution is applied by updating the bed field $b(x, y, t)$; no additional depth-equation mass source is added in the conserved-depth formulation.

5.11 Initial, Boundary and Interface Conditions

5.12 Model Coupling Requirements

5.13 Sufficient Conditions for Validity

5.14 Candidate Approaches

5.15 Critical Comparison

5.16 Assumptions and Applicability

5.17 Limitations and Failure Conditions

5.18 Project-Specific Conclusions

For the regional model, the source-term research gate is closed at specification level. Remaining work is numerical verification of the well-balanced bed-source treatment, validation of the selected earthquake source against observations, and selection of data values for roughness, bathymetry, and source timing. These are not changes to the regional mathematical model unless the validation evidence rejects the specified formulation.

5.19 Selected Approach and Rationale

The selected regional source approach is:

- bed-slope forcing with hydrostatic reconstruction and matching bed-source discretisation;
- Manning bottom friction, advanced by a split semi-implicit update in the numerical method;
- dynamic moving-bed earthquake source, $b = b_0 + \sum_m \Delta b_m R_m(t)$;
- sponge-layer relaxation at artificial ocean boundaries where required by domain truncation;
- Coriolis and explicit lateral eddy viscosity disabled in the baseline, retained only as sensitivity options.

The selected local closure approach is:

- high-Reynolds-number URANS as the operational local hydrodynamic model;
- $k-\omega$ SST as the baseline eddy-viscosity closure;
- LES retained for selected high-fidelity comparison cases;
- molecular-viscosity-only Navier–Stokes retained only as a diagnostic calculation;
- porous, compressible-air, sediment, scour and structural damage source terms deferred until their activation gates are met.

5.20 Unresolved Decisions

The mathematical form of the regional source terms and the local URANS baseline closure are no longer unresolved. The open items are data, verification and lower-level closure-implementation items:

- final Manning coefficient fields;
- final static bathymetry/topography dataset;
- final fault-source dataset and rupture-time functions;
- lake-at-rest and moving-bottom verification evidence;
- decision on whether optional Coriolis sensitivity is required;
- local wall-function constants and roughness treatment for the selected case-study materials;
- final LES sub-grid closure and comparison-case set.

5.21 Requirements and Handoffs

‘RES-NUM-SRC’ shall implement the source terms consistently with the finite-volume face fluxes, preserve still-water balance over non-flat bathymetry, and report any loss of positivity at wet/dry fronts. ‘RES-DAT-GEO’ and ‘RES-DAT-SRC’ shall provide the static bed, coordinate/datum transform, and fault-source data required by the source definitions.

‘RES-NUM-SPEC’ shall discretise Equations (5.5) and (5.6) after the pressure-corrected velocity and face flux have been accepted for the time step. ‘RES-PHY-SCL’ shall justify the high-Reynolds-number regime, and ‘RES-VER-MET’ shall prevent free-surface agreement alone from being treated as closure validation.

5.22 Deliverable Completion Record

6 RES-MOD-IBC — Initial, Boundary and Interface Condition Models

6.1 Purpose and Scope

6.2 Research Questions

6.3 Terminology and Definitions

6.4 Literature Evidence

Higuera, Lara, and Losada [8] is retained as evidence that Navier–Stokes numerical wave tanks require compatible wave generation, active absorption and relaxation/open-boundary treatments. Source role: ‘3D-WAVEBC’. Project conclusion: the local inlet, outlet and lateral ocean cuts shall be treated as wave-boundary components rather than generic zero-gradient faces. Limitation: exact relaxation-zone lengths and damping profiles remain numerical and validation choices.

Takase et al. [2] and Romano et al. [3] support independent regional and local domains connected through interface state transfer. Source role: ‘3D-COUPPING’. Project conclusion: local inlet histories shall be reconstructable from regional water depth, surface elevation and discharge histories. Limitation: the initial depth-uniform hydrostatic lift is not a unique reconstruction of the three-dimensional state.

6.5 Governing Relations and Quantitative Definitions

6.6 System Boundary

6.7 State Variables and Parameters

6.8 Continuous Governing Model

6.9 Source Terms and Closures

6.10 Initial, Boundary and Interface Conditions

The regional two-dimensional model uses a horizontal depth-averaged state. Initial conditions therefore consist of the cellwise water depth, depth-integrated discharges, static bed, and any time-dependent earthquake bed increments. Cross-sectional views are diagnostic outputs derived from the horizontal solution and shall not be used as separate one-dimensional solved domains.

Artificial ocean boundaries are represented by boundary states $\mathbf{U}_B$ supplied to the numerical flux. The baseline mathematical boundary roles are:

Boundary or interface	Regional treatment	Status
Offshore artificial boundary	Characteristic or radiation open boundary	Established at specification level; validation pending
Lateral artificial ocean cuts	Radiation or open boundary, with sponge damping where required	Established at specification level; domain-specific tuning pending
Physical coastline	Topography plus wetting–drying, not an external wall boundary	Established at specification level
True wall or resolved structure	Reflective boundary state	Established for walls; final defence representation remains RES-GEO/RES-MOD-MUL work
Regional-to-local interface	Conservative transfer of h , η , q_x , q_y and derived boundary histories	Coupling algorithm deferred

Table 3. Regional boundary and interface condition roles

6.10.1 Local Three-Dimensional Boundary Roles

The local boundary partition is defined in Equation (3.2). The continuous local boundary roles are:

For one-way replay the regional model supplies h_R , η_R , $q_{x,R}$ and $q_{y,R}$ on the interface. The depth-averaged velocities are:

Boundary	Continuous role	Project status
Γ_{in}	Regional-to-local incident-wave forcing	Selected for one-way replay; also used by simultaneous coupling
Γ_{out}	Downstream open or radiation boundary with absorption where needed	Selected as a boundary role; exact formula and relaxation length remain numerical choices
Γ_{side}	Lateral open-ocean cuts	Selected as open/radiation boundaries, not physical walls
Γ_{top}	Open atmosphere	Selected as an open pressure/reference-air boundary
Γ_{T}	Fixed terrain and bathymetry	Selected as a no-slip wall with wall-model details owned by ‘RES-NUM-IBC’
Γ_{B}	Fixed rigid barrier surface	Selected for baseline hydrodynamics; structural motion deferred to activated FSI

Table 4. Continuous local three-dimensional boundary-condition roles

$$u_{\text{R}} = \frac{q_{x,\text{R}}}{h_{\text{R}}} \quad (6.1)$$

$$v_{\text{R}} = \frac{q_{y,\text{R}}}{h_{\text{R}}} \quad (6.2)$$

The baseline local inlet reconstruction is:

$$\alpha_{\text{in}} = \mathcal{H}(\eta_{\text{R}} - z) \quad (6.3)$$

$$\mathbf{u}_{\text{in}} = [u_{\text{R}}, v_{\text{R}}, 0]^{\text{T}} \quad (6.4)$$

$$p_{\text{in}} = \rho_{\text{w}} g \max(\eta_{\text{R}} - z, 0) \quad (6.5)$$

where $\mathcal{H}$ is a phase-indicator reconstruction. This is a depth-uniform, hydrostatic lift and is valid only where the interface is upstream of strong local vertical acceleration, breaking and barrier-induced separation.

On the fixed terrain and barrier surfaces the baseline velocity condition is:

$$\mathbf{u}_{\text{f}} = \mathbf{0} \quad \text{on} \quad \Gamma_{\text{T}} \cup \Gamma_{\text{B}} \quad (6.6)$$

The no-slip condition defines the continuous wall role. The high- Reynolds-number wall-function approximation used by the discrete model belongs to ‘RES-NUM-IBC’ and shall not be mistaken for a change in the continuous boundary condition.

6.11 Model Coupling Requirements

The local boundary histories shall remain event-independent in formulation. A Tohoku case, or any later event, changes the supplied time series and terrain data, not the mathematical boundary roles in Table 4.

6.12 Sufficient Conditions for Validity

6.13 Candidate Approaches

6.14 Critical Comparison

6.15 Assumptions and Applicability

6.16 Limitations and Failure Conditions

6.17 Project-Specific Conclusions

6.18 Selected Approach and Rationale

The selected regional condition model is a face-state boundary formulation. Boundary conditions act during every residual evaluation, not as a post-step correction. This preserves consistency with the finite-volume conservation update and lets open, reflective, sponge and interface boundaries share one mathematical mechanism.

6.19 Unresolved Decisions

The regional boundary-condition roles are established, but the following remain open:

- final QGIS/geospatial domain lock and artificial-boundary placement;
- sponge-layer width and damping scale after domain selection;
- final regional-to-local transfer operators;
- final representation of documented defences as topography, wall faces, holes, porous interfaces, or local 3D extraction regions.
- local inlet relaxation treatment and pressure compatibility under strong reflection;
- local wall-function and roughness parameters for terrain and barrier surfaces;
- local outlet and side-boundary absorption settings.

6.20 Requirements and Handoffs

‘RES-NUM-IBC’ shall express each boundary role as a discrete boundary state or transfer operator. ‘RES-DAT-GEO’ shall provide the final domain, coastline, datum and coordinate transforms before boundary locations are locked.

‘RES-VER-FRM’ and ‘RES-VER-MET’ shall test whether boundary reflections or inlet reconstruction errors alter run-up, overtopping, pressure, force, moment or candidate ranking.

6.21 Deliverable Completion Record

7 RES-MOD-LRG — Large-Domain Representation and Model Reduction

7.1 Purpose and Scope

7.2 Research Questions

7.3 Terminology and Definitions

7.4 Literature Evidence

7.5 Governing Relations and Quantitative Definitions

7.6 System Boundary

7.7 State Variables and Parameters

7.8 Continuous Governing Model

The selected large-domain regional representation is the horizontal two-dimensional depth-averaged NLSWE model defined in ‘RES-MOD-EQS’ and consolidated in ‘RES-MOD-SPEC’. It represents Tōhoku-scale generation, propagation, nearshore transformation and inundation without applying local three-dimensional CFD resolution across the whole domain.

7.9 Source Terms and Closures

7.10 Initial, Boundary and Interface Conditions

7.11 Model Coupling Requirements

7.12 Sufficient Conditions for Validity

7.13 Candidate Approaches

7.14 Critical Comparison

7.15 Assumptions and Applicability

7.16 Limitations and Failure Conditions

7.17 Project-Specific Conclusions

7.18 Selected Approach and Rationale

Use the regional 2D NLSWE finite-volume solver as the large-domain model. Preserve local three-dimensional modelling only where structure-scale pressure, overtopping, local air–water interface topology or FSI requires vertically resolved flow.

7.19 Unresolved Decisions

Open large-domain decisions are the QGIS/geospatial domain extent, artificial-boundary placement, sponge-layer placement, local 3D extraction region and final mesh-refinement levels.

7.20 Requirements and Handoffs

7.21 Deliverable Completion Record

8 RES-MOD-MUL — Multidimensional and Multifidelity Coupling Strategy

8.1 Purpose and Scope

This Deliverable defines how the regional two-dimensional NLSWE model and the local three-dimensional URANS–VOF model are connected as continuous model components. It records coupling roles, validity conditions and handoffs, while leaving discrete interpolation, projection, residuals and software scheduling to ‘RES-NUM’ and Software.

8.2 Research Questions

The coupling decision asks:

1. When is one-way regional-to-local replay sufficient for candidate screening?
2. When must the local three-dimensional model exchange feedback with a retained regional buffer?
3. Which state variables must be lifted and restricted between depth-averaged and three-dimensional descriptions?
4. Which validation tests decide that the cheaper coupling mode is acceptable?

8.3 Terminology and Definitions

8.4 Literature Evidence

8.4.1 Romano et al. (2025)

Romano et al. [3] developed a sequential coupling strategy for a landslide-generated tsunami in complex fjord geometry. A three-dimensional model was retained through the near-field region in which source generation, strong nonlinearity, confinement, inundation, and topobathymetric transformation remained important. Its output was then transferred to a computationally cheaper two-dimensional model for subsequent propagation.

The paper is retained for its treatment of model selection, interface placement, and information transfer rather than for its landslide-source formulation. The present project uses an earthquake-derived initial condition and a nonlinear shallow-water regional model, but the same modelling principle applies: the transition between models shall occur only after the physics omitted by the receiving model have become sufficiently weak.

Romano et al. also varied the coupling-boundary position. Their results demonstrated that geometric distance from the source was not, by itself, a sufficient placement criterion. A reduced model could only reproduce the subsequent propagation satisfactorily after the wave had completed the strongly nonlinear and geometrically controlled transformations that required the three-dimensional representation.

8.4.2 Takase et al. (2016)

Takase et al. [2] coupled a two-dimensional shallow-water model to a local three-dimensional free-surface Navier–Stokes model using independently generated unstructured meshes. Multiple-point constraints and transmission relations were used to maintain compatibility of velocity and pressure across non-conforming interfaces.

The paper is retained as evidence that the two-dimensional and three-dimensional regions may remain independently discretised while exchanging the physical variables required for a consistent hybrid solution. Its finite-element constraint formulation is not adopted directly, as the regional solver developed in this project will use a finite-volume discretisation.

8.5 Governing Relations and Quantitative Definitions

8.6 System Boundary

The complete hydrodynamic system shall be decomposed into:

1. a regional two-dimensional domain in which the depth-averaged shallow-water assumptions remain defensible;
2. a local three-dimensional domain in which vertical motion, free-surface deformation, overtopping, separation, impact, and detailed barrier geometry must be resolved;
3. an interface layer responsible for lifting, restricting, synchronising, and conserving information between the two model representations.

The transition shall occur before local three-dimensional processes invalidate the depth-averaged representation. The interface location shall remain configurable rather than being treated as a permanently fixed coordinate.

8.7 State Variables and Parameters

The regional state may be represented by:

$$\mathbf{U}_{2D} = [h, h\bar{u}, h\bar{v}]^T \quad (8.1)$$

where h is total water depth and $\bar{u}$ and $\bar{v}$ are depth-averaged horizontal velocities.

The local state may be represented by:

$$\mathbf{U}_{3D} = [\mathbf{u}, p, \alpha]^T \quad (8.2)$$

where $\mathbf{u}$ is the three-dimensional velocity field, p is pressure, and α denotes the selected free-surface or phase representation.

8.8 Continuous Governing Model

8.9 Source Terms and Closures

8.10 Initial, Boundary and Interface Conditions

8.11 Model Coupling Requirements

A lifting operator shall convert the depth-averaged regional state into a three-dimensional interface state:

$$\mathcal{L}_{2 \rightarrow 3}(\vec{U}_{2D}) = \vec{U}_{3D,\Gamma} \quad (8.3)$$

A restriction operator shall convert the three-dimensional interface state or interface fluxes back into a depth-integrated regional state:

$$\mathcal{R}_{3 \rightarrow 2} \left(\vec{U}_{3D,\Gamma} \right) = \vec{U}_{2D,\Gamma} \quad (8.4)$$

The operators shall preserve, within the specified numerical tolerances:

- free-surface compatibility;
- normal mass flux;
- normal and tangential momentum transfer;
- hydrostatic pressure consistency where the shallow-water assumptions remain applicable;
- spatial mapping between non-conforming meshes;
- temporal consistency between different solver time steps;
- propagation of outgoing and reflected disturbances.

The lifting operator shall initialise the local three-dimensional state from the regional solution. It shall not be interpreted as recovering a unique three-dimensional field from depth-averaged information. The local solver shall develop the resolved vertical and vortical structure after initialisation.

8.12 Sufficient Conditions for Validity

The one-way replay mode shall be considered sufficient only when:

- the interface lies outside the region of material three-dimensional flow;
- the barrier does not materially alter the incident state at the selected interface;
- outgoing reflected waves are not artificially trapped or suppressed;
- transferred mass and momentum remain within the specified imbalance tolerances;
- quantities of interest remain insensitive to reasonable upstream movement of the interface;
- comparison with the simultaneously coupled model remains within the defined fidelity tolerances.

Where these conditions are not satisfied, the simultaneously coupled mode shall be used.

An interface shall be placed only where the state can be represented adequately by the governing assumptions of the receiving model. Its location shall therefore account for:

- wave nonlinearity;

-
- vertical acceleration and velocity structure;
 - hydrostatic-pressure validity;
 - wave breaking and free-surface fragmentation;
 - geometric confinement;
 - abrupt bathymetric contraction or expansion;
 - local resonance or reflection;
 - barrier-induced wake and recirculation;
 - inundation and moving-shoreline behaviour;
 - the frequency content retained by the receiving model.

The upstream two-dimensional-to-three-dimensional interface shall be positioned before processes requiring the three-dimensional model become dominant. The downstream three-dimensional-to-two-dimensional interface shall be positioned after those processes have decayed sufficiently.

The interface shall not be accepted solely from qualitative inspection. Its suitability shall be demonstrated by moving the interface and showing convergence of the quantities of interest.

8.13 Candidate Approaches

Four operational modes shall be supported conceptually.

8.13.1 Standalone Two-Dimensional Mode

The regional model shall simulate earthquake-generated tsunami formation, offshore propagation, bathymetric transformation, shoaling, and regional inundation while its assumptions remain valid.

8.13.2 Standalone Three-Dimensional Mode

The local model shall support prescribed analytical, experimental, or numerically reconstructed inflow conditions for component verification and controlled barrier studies.

8.13.3 One-Way Replay Mode

A saved regional checkpoint or stored interface history shall supply the incident state to the local three-dimensional model. The three-dimensional solution shall not modify the regional solution in this mode. For each environmental scenario, the full regional simulation may be advanced once to a selected pre-impact time t_s . The saved checkpoint shall contain the regional restart state, interface variables, solver time, and all numerical metadata required for deterministic reconstruction.

8.13.4 Simultaneously Coupled Mode

The retained regional subdomain and local three-dimensional domain shall advance over the same physical time interval while exchanging information at defined coupling points. This mode shall permit barrier-generated reflection and other local disturbances to influence the retained regional solution. The coupled model may remain partitioned, with the two solvers executed as independent modules. Monolithic assembly of both equation systems is not required provided that the partitioned exchange converges and satisfies the prescribed conservation and compatibility conditions.

The model architecture shall permit different spatial arrangements of the two-dimensional and three-dimensional domains.

A conventional upstream transition may be represented as:

$$\Omega_{2D}^{up} \xrightarrow{\mathcal{L}_{2 \rightarrow 3}} \Omega_{3D} \quad (8.5)$$

where the regional two-dimensional state is lifted into a local three-dimensional state before the flow enters the region of strong vertical motion or barrier interaction.

A downstream transition may be represented as:

$$\Omega_{3D} \xrightarrow{\mathcal{R}_{3 \rightarrow 2}} \Omega_{2D}^{down} \quad (8.6)$$

where the local three-dimensional result is restricted to a depth-integrated state after the barrier-induced wake, vertical shear, free-surface fragmentation, and other three-dimensional processes have decayed sufficiently.

The upstream and downstream interfaces need not occupy the same location. They may use different transfer operators and validity criteria, reflecting the different physical states entering and leaving the local three-dimensional domain.

8.14 Multidimensional and Multifidelity Coupling Strategy

The baseline workflow shall separate the expensive source-to-coast propagation from the repeatedly executed barrier-interaction stage:

1. generate the tsunami using an earthquake-derived initial condition;
2. advance the regional two-dimensional model towards the selected coastal region;
3. save a complete pre-impact checkpoint at time t_s ;
4. restore the checkpoint for each candidate barrier geometry;
5. initialise the local three-dimensional model through $\mathcal{L}_{2 \rightarrow 3}$;
6. execute either the one-way replay mode or the simultaneously coupled mode;

-
7. calculate the hydrodynamic quantities of interest;
 8. compare, rank, or modify the candidate geometry.

The saved state shall include a retained two-dimensional buffer upstream of the local three-dimensional domain when reflected-wave feedback is to be represented. The upstream edge of this buffer shall receive the incident-wave forcing while permitting outgoing reflected disturbances to leave without material numerical contamination.

8.15 Critical Comparison

The one-way replay model is computationally cheaper and permits the same regional incident state to be reused across many candidate geometries. It neglects feedback from the local three-dimensional domain into the regional solution. The simultaneously coupled model represents this feedback but requires continued execution of the retained two-dimensional domain, bidirectional interface transfer, temporal synchronisation, and additional convergence controls. Neither mode shall be selected universally. Their difference shall be quantified using identical initial checkpoints, geometries, interface locations, meshes, physical parameters, and output definitions.

Romano et al. used a three-dimensional granular-landslide and free-surface model in the near field and a linear mild-slope model for far-field propagation. Those selections were appropriate to their source mechanism, complex fjord geometry, and far-field modelling objective. They are not transferred directly into the present earthquake-tsunami framework.

The following elements are adopted:

- assign each spatial region the least expensive model that retains the locally important physics;
- place the interface according to model validity rather than distance alone;
- assess several candidate interface locations;
- retain the high-fidelity model through geometric bottlenecks, strong confinement, nonlinear transformation, and other processes outside the reduced model's applicability;
- permit a saved high-fidelity history to initialise a cheaper downstream model;
- preserve the temporal and spectral content of the transferred wave;
- separate the physical coupling strategy from the particular source-generation model.

The following elements are not adopted directly:

- the granular-landslide source, as the primary benchmark is generated through earthquake-induced seabed deformation;
- the linear mild-slope equation, as the regional model must support nonlinear propagation, wetting–drying, and inundation;

- a free-surface-only transfer, as a nonlinear shallow-water receiver requires water depth and depth-integrated discharge;
- frequency-domain propagation as the primary execution method, as the planned coupling architecture is time-domain and restartable;
- a fixed shoreline, as coastal inundation forms part of the benchmark problem.

These choices are omitted as case-specific implementations rather than treated as invalid methods. Their role in the paper clarifies why the coupling variables and reduced model must be selected according to the receiving model's governing equations and intended outputs.

8.16 Assumptions and Applicability

8.17 Limitations and Failure Conditions

8.18 Project-Specific Conclusions

The project shall adopt a modular and restartable two-dimensional– three-dimensional architecture rather than one fixed hybrid configuration. The physical solvers, checkpoint system, lifting operator, restriction operator, and coupling controller shall remain separable components.

The one-way replay and simultaneously coupled models shall both be retained. The one-way model shall provide an efficient candidate-screening path, while the coupled model shall provide the reference treatment when barrier-generated reflections or other upstream effects are material.

The selection between the two modes shall be evidence-based. A direct comparison shall determine whether the additional coupling fidelity causes a substantial change in the engineering quantities used to evaluate a barrier.

Romano et al. support a modular interpretation of the coupled model in which the governing formulation may change more than once across the domain. The project shall therefore not be restricted to a single two-dimensional-to-three-dimensional transition.

The target architecture shall support:

$$\Omega_{2D}^{\text{up}} \xrightarrow{\mathcal{L}_{2 \rightarrow 3}} \Omega_{3D} \xrightarrow{\mathcal{R}_{3 \rightarrow 2}} \Omega_{2D}^{\text{down}} \quad (8.7)$$

where either two-dimensional region may be omitted when unnecessary. The central three-dimensional region shall extend far enough upstream and downstream to contain all processes that cannot be represented reliably by the regional depth-averaged equations.

The same operators shall support the saved one-way replay model and the simultaneously coupled model. Differences between those execution modes shall be determined through direct comparison rather than presumed from the coupling direction alone.

The Tōhoku hindcast literature provides further evidence that the complete event should be represented as a chain of purpose-specific models rather than by one uniformly complex formulation.

The model chain may include:

$$\mathcal{M}_{\text{source}} \rightarrow \mathcal{M}_{\text{regional}} \rightarrow \mathcal{M}_{\text{nearshore}} \rightarrow \mathcal{M}_{\text{local-3D}} \quad (8.8)$$

Each transition shall be justified by:

- a change in the dominant physical processes;
- a change in the validity of the governing assumptions;
- a requirement for increased spatial or dimensional resolution;
- a demonstrated reduction in computational cost relative to applying the highest-fidelity model everywhere.

A limitation of one model shall not be interpreted automatically as a limitation of the complete architecture. Where appropriate, the model shall be terminated before its assumptions fail and its state shall be transferred into a formulation capable of representing the subsequent physics.

The reviewed regional hindcasts support this modular principle but do not replace the dedicated two-dimensional–three-dimensional coupling evidence assigned elsewhere in this Deliverable.

8.19 Selected Approach and Rationale

The selected baseline execution mode is one-way replay:

regional 2D NLSWE checkpoint and interface history → local 3D URANS–VOF
tsunami–barrier interaction simulation

Source role: ‘3D-COUPLING’. Supporting sources: Takase et al. [2] and Romano et al. [3]. Project-specific conclusion: one regional event reconstruction can drive repeated local barrier simulations for efficient candidate screening. Limitation: the one-way branch is not accepted when barrier-generated reflection, blocking, resonance or downstream feedback materially changes quantities of interest.

The higher-fidelity mode is simultaneous coupling with a retained two-dimensional buffer and a local three-dimensional domain. This mode shall be activated when direct comparison shows that one-way replay changes safety-critical outputs or candidate ranking beyond the accepted tolerance. The mathematical coupling architecture is event-independent; a selected event changes source, bathymetry, topography and boundary histories rather than the coupling formulation.

For the active Tōhoku baseline, the replay histories are extracted from the Kamaishi and Sendai corridors defined in ‘RES-DAT-SCN’ and consolidated in ‘RES-DAT-CAS’. Corridor construction is therefore a data/model interface: it selects the inlet section, buffer region, terrain subset and observation bundle used by the coupling test. It is not a separate physics model and shall not be adjusted by hand to improve a local 3D result.

8.20 Unresolved Decisions

Open coupling decisions are the exact lifting and restriction operators, temporal interpolation method, interface-location study, feedback residuals, convergence criteria and acceptance tolerances for selecting between replay and simultaneous coupling.

8.21 Requirements and Handoffs

The Software workstream shall implement:

- deterministic saving and restoration of complete regional checkpoints;
- independently executable two-dimensional and three-dimensional solver modules;
- lifting and restriction operators;
- configurable interface locations;
- manifest-linked corridor and replay-boundary identifiers;
- one-way and bidirectional coupling modes;
- configurable coupling intervals and temporal interpolation;
- conservation and compatibility diagnostics;
- concurrent execution of comparison branches;
- a synchronisation point that prevents comparison from proceeding until both branches have completed.

The one-way and coupled comparison branches shall operate on separate copies of an immutable checkpoint and shall not share mutable solver state. Compute resources shall be partitioned explicitly to avoid thread or process oversubscription. The Verification workstream shall define the fidelity metrics, acceptance tolerances, interface-location study, and conservation tests used to approve either execution mode.

8.22 Deliverable Completion Record

9 RES-MOD-ASM — Assumptions, Applicability and Model Limitations

9.1 Purpose and Scope

This Deliverable records the assumptions under which the local three-dimensional URANS–VOF hydrodynamic model may be used as the baseline tsunami–barrier interaction model. It also identifies physical regimes that require deferred extensions before claims can be made.

9.2 Research Questions

The local-model applicability questions are:

1. Which physical effects are included in the baseline local hydrodynamic model?
2. Which effects are deliberately deferred rather than assumed negligible?
3. Which candidate geometries trigger porous, compressible-air, structural-failure, scour or sediment extensions?
4. Which claims may be made before those extensions and validation data exist?

9.3 Terminology and Definitions

9.4 Literature Evidence

9.4.1 del Jesus et al. (2012)

del Jesus, Lara, and Losada [9] demonstrated that unresolved porous barrier material requires volume averaging and additional porous-matrix resistance. This formulation is relevant to rubble mounds, densely packed armour units, aggregate-filled structures, and other media whose pore geometry is smaller than the resolved computational scale.

9.4.2 Istrati and Buckle (2019)

Istrati and Buckle [10] showed that enclosed or poorly ventilated air volumes can alter local pressure, uplift, and overturning moment. The study does not imply that compressible flow is required for every tsunami-impact calculation. It identifies a geometry-dependent failure regime for the incompressible-air assumption.

9.5 Applicability Boundary of the Local Three-Dimensional Model

The selected incompressible two-phase Navier–Stokes formulation is the baseline local model for rigid, impermeable, and well-ventilated barriers. Several alternative or extended formulations were considered but are not included in the initial model.

Formulation	Reason for consideration	Project decision
Compressible two-phase Navier–Stokes	Represents trapped-gas compression, acoustic effects, and compressibility	Deferred while the design space remains open and ventilated; activated only when a trapped-air screening gate fails
VARANS	Represents unresolved internal porous flow through porosity and matrix-resistance closures	Deferred while the initial barrier is impermeable or contains explicitly resolved openings
Three-dimensional nonhydrostatic multilayer equations	Provides some vertical resolution at lower dimensional complexity	Not selected owing to limited evidence for local pressure, traction, and general moving-boundary FSI
Three-dimensional hydrostatic multilayer equations	Provides vertically layered velocity and density fields	Rejected for impulsive local impact as the pressure field remains hydrostatic
Inviscid Euler or potential-flow equations	Can represent selected non-breaking wave and body-interaction problems	Rejected for the operational impact model as viscosity, turbulent separation, wake formation, and dissipative mechanisms are central outputs
Molecular-viscosity-only Navier–Stokes	Retains the parent conservation laws without an explicit unresolved-turbulence closure	Retained only as a diagnostic case; not accepted as the high-Reynolds-number operational model
SPH or MPS	Handles highly deformed free surfaces through mesh-free numerical formulations	Deferred to numerical-method selection as these are discretisation frameworks rather than alternative conservation laws

Table 5. Alternative and extended formulations considered for the local three-dimensional model

9.5.1 Considered and Deferred Formulations

9.6 Governing Relations and Quantitative Definitions

9.7 System Boundary

9.8 State Variables and Parameters

9.9 Continuous Governing Model

9.10 Source Terms and Closures

9.11 Initial, Boundary and Interface Conditions

9.11.1 Initial Local-Model Assumptions

The initial local model adopts the following assumptions:

1. water and air are Newtonian fluids;
2. both phases are incompressible;
3. the phases are immiscible;
4. heat transfer, evaporation, condensation, and phase change are negligible;
5. the barrier is initially rigid and impermeable;
6. candidate geometries are sufficiently ventilated to prevent material trapped-air compression;
7. seabed morphology is fixed during one hydrodynamic simulation;
8. debris, sediment transport, scour, cavitation, and structural fracture are excluded from the baseline model;
9. turbulence effects are represented through an attached URANS, LES, or hybrid closure;
10. structural pressure and traction are retained even when the structure is fixed.

These assumptions constrain the initial design space rather than assert that the excluded phenomena are universally negligible.

9.11.2 Porous-Medium Activation Gate

A porous-region model shall be considered where:

$$\ell_{\text{pore}} \ll \Delta_{\text{local}} \tag{9.1}$$

where ℓ_{pore} is a representative pore scale and Δ_{local} is the local resolved mesh scale.

Where individual slots, channels, perforations, or openings are design variables and can be resolved explicitly, the ordinary Navier–Stokes equations shall remain active.

Where the internal porous structure cannot be resolved, a local resistance term may be attached:

$$\mathbf{f}_{\text{por}} = - \left[a \mathbf{u}_f + b \|\mathbf{u}_f\| \mathbf{u}_f + c \frac{\partial \mathbf{u}_f}{\partial t} \right] \quad (9.2)$$

The coefficients a , b , and c shall be calibrated for the selected material and shall not be transferred directly from unrelated porous structures.

9.11.3 Compressible-Air Activation Gate

A candidate geometry shall be flagged for compressible-air assessment where:

1. an initially connected air region becomes isolated during impact;
2. the trapped volume persists for a material fraction of the loading interval;
3. water intrusion causes a material reduction in gas volume;
4. the venting timescale is comparable with or greater than the impact timescale;
5. cavity pressure contributes materially to the resultant force or overturning moment.

A preliminary screening estimate may use:

$$p_a V_a^n = p_{a,0} V_{a,0}^n \quad (9.3)$$

where n lies between the isothermal and adiabatic limits.

Where the predicted gas-pressure increment is comparable with the local hydrodynamic pressure, the geometry shall be ventilated, excluded from the baseline design space, or assessed using a compressible-gas extension.

9.11.4 Current Validity Statement

The selected local equation set is applicable to:

- rigid or weakly deforming barriers;
- impermeable barriers;
- explicitly resolved openings and flow passages;

-
- open or adequately ventilated geometries;
 - water–air free-surface impact without material gas compression;
 - fixed-bed hydrodynamic evaluation;
 - one-way structural load transfer;
 - two-way FSI where the structural model and moving-interface treatment are activated.

The baseline formulation shall not be claimed to reproduce unresolved porous flow, trapped-air compression, debris impact, movable sediment, scour, cavitation, fracture, or progressive structural failure without the corresponding additional physical model.

9.12 Model Coupling Requirements

9.13 Sufficient Conditions for Validity

9.14 Candidate Approaches

9.15 Critical Comparison

9.16 Assumptions and Applicability

The local model assumes high-Reynolds-number water–air impact flow around a fixed or weakly moving defence geometry. The operational closure is URANS $k-\omega$ SST; LES is retained for selected high-fidelity comparison cases. Source roles: ‘3D-SST’, ‘3D-RANSVOF’, ‘3D-DEFERRED’. Supporting sources: Menter [6], Kozelkov et al. [7], del Jesus, Lara, and Losada [9] and Istrati and Buckle [10]. Project-specific conclusion: turbulence, free-surface deformation and hydrodynamic loading are baseline phenomena, while unresolved porous media and trapped-air compressibility are activation-gated extensions.

9.17 Limitations and Failure Conditions

The baseline model ceases to be sufficient when a quantity of interest depends materially on:

- unresolved flow through porous material;
- isolated trapped-air pockets with material compression;
- sediment transport, scour or bed morphology change;
- debris impact or floating-body collision;
- structural fracture, progressive damage or detached debris;
- cavitation or compressible acoustic effects;
- certification-level design factors not covered by the validation evidence.

In those regimes the result may be used only as a hydrodynamic screening calculation unless the missing physical model is added and validated.

9.18 Project-Specific Conclusions

The local 3D model is valid as a baseline high-Reynolds-number hydrodynamic tsunami-barrier interaction model for run-up, overtopping, free-surface deformation, pressure, force, moment, and comparative barrier-performance analysis. It is not valid yet for structural failure, scour, deformable barriers, detailed aeration, sediment transport, or final design certification.

9.19 Selected Approach and Rationale

The selected baseline assumption set is:

incompressible immiscible water–air flow, VOF free surface, high-Reynolds-number URANS $k-\omega$ SST turbulence, fixed terrain, fixed rigid impermeable and ventilated barrier geometry, and hydrodynamic load extraction without baseline structural failure, scour, sediment or trapped-air compression

9.20 Unresolved Decisions

Unresolved applicability details are the quantitative trapped-air screening threshold, rough-wall turbulence parameters, porous-extension activation threshold, deformation threshold for activating FSI, and case-study-specific validation tolerance for treating the barrier as rigid.

9.21 Requirements and Handoffs

‘RES-GEO’ shall flag candidate geometries that violate the baseline ventilation, porosity or rigidity assumptions. ‘RES-VER-MET’ shall define the validation checks that decide whether the hydrodynamic baseline is acceptable for candidate screening. ‘RES-STR-FSI’ shall own any structural-feedback or failure extension.

9.22 Deliverable Completion Record

10 RES-MOD-SPEC — Integrated Mathematical Model Specification

10.1 Purpose and Scope

This Deliverable consolidates the continuous mathematical model for the regional tsunami solver and the local three-dimensional hydrodynamic tsunami–barrier model. The regional 2D

NLSWE mathematical model and the local 3D URANS–VOF hydrodynamic model are complete at research-specification level. Remaining work belongs to geospatial/domain selection, validation, implementation, detailed coupling operators and activated FSI or structural-failure extensions.

10.2 Research Questions

The regional-model gate answers:

1. Which horizontal governing model represents regional tsunami propagation and inundation?
2. Which state variables, bed convention, source terms and condition roles define the mathematical model?
3. Which decisions are established and which downstream workstreams remain open?
4. Which continuous local model is used for pressure, force, moment, overtopping and barrier-performance comparison?
5. Which local physics are baseline assumptions and which are deferred extensions?

10.3 Terminology and Definitions

10.4 Literature Evidence

10.5 Governing Relations and Quantitative Definitions

10.6 System Boundary

The regional solver is a horizontal, depth-averaged, cell-centred finite-volume model for the two-dimensional nonlinear shallow-water equations. Its independent solved coordinates are (x, y, t) . Vertical information is carried through scalar fields $b(x, y, t)$, $\eta(x, y, t)$ and $h(x, y, t) = \eta - b$. Cross-sections are diagnostic visualisations extracted from the horizontal solution rather than separately solved cross-sectional models.

10.7 State Variables and Parameters

The conserved regional state is Equation (4.1). The primitive state is Equation (4.4). The bed convention is positive upward, with $\eta = h + b$ and $h = \eta - b$.

10.8 Continuous Governing Model

The established regional governing model is the nonlinear shallow-water balance law in Equation (4.6), with Cartesian fluxes in Equations (4.7) and (4.8). The model supports wetting and

drying through the numerical method; it does not introduce a regional volume-of-fluid phase variable.

The established local governing model is the three-dimensional nonhydrostatic incompressible immiscible two-phase Navier–Stokes system with VOF phase transport: Equations (4.16), (4.17) and (4.20). The local state is Equation (4.28); the operational turbulence closure is URANS $k-\omega$ SST.

10.9 Source Terms and Closures

The baseline regional source terms are bed slope, Manning friction and open-boundary sponge damping. Dynamic earthquake forcing is represented as a time-dependent bed field, Equation (4.13). Since h is the conserved depth variable, no separate earthquake mass source is added to the depth equation.

Coriolis and explicit lateral eddy viscosity remain disabled in the baseline and may be activated only as later sensitivity studies.

The local effective viscosity is defined by Equation (4.23), with μ_t supplied by the selected URANS $k-\omega$ SST closure in Equations (5.5) to (5.7). LES is retained only for high-fidelity comparison cases, not as the baseline repeated-screening model.

10.10 Initial, Boundary and Interface Conditions

The regional condition roles are those in Table 3: characteristic or radiation open boundaries for artificial ocean boundaries, optional sponge damping for artificial cuts, coastline treatment by topography and wetting–drying, reflective states for true walls, and conservative regional-to-local transfer data for the later interface workstream.

The local condition roles are those in Table 4: regional-to-local inlet forcing, open or radiation downstream and lateral boundaries, an open atmospheric top, fixed no-slip terrain and a fixed no-slip barrier for the hydrodynamic baseline.

10.11 Model Coupling Requirements

The regional model supplies h , η , q_x , q_y , derived velocities, boundary histories and checkpoint data to the local 3D/interface workstream. The local 3D model receives those histories through the baseline one-way replay mode defined in ‘RES-MOD-MUL’. Simultaneous coupling remains required when feedback or reflection materially changes the quantities of interest.

10.12 Sufficient Conditions for Validity

The regional model is valid only where the depth-averaged, hydrostatic, long-wave assumptions are acceptable for the selected outputs. It is not the model for local three-dimensional impact pressure, vertical velocity structure, resolved overtopping jets, local air–water interface topology or structural FSI.

The local model is valid as a baseline hydrodynamic model where the barrier is fixed or weakly moving, impermeable or explicitly resolved, and sufficiently ventilated that trapped-air compression is not material. It is not a model for unresolved porous media, scour, sediment transport, debris impact, structural failure or final design certification without additional physics and validation.

10.13 Candidate Approaches

10.14 Critical Comparison

10.15 Assumptions and Applicability

10.16 Limitations and Failure Conditions

10.17 Project-Specific Conclusions

The regional 2D NLSWE formulation is now established at research-specification level. It shall be treated as the initial mathematical baseline for implementation and verification planning. Future changes to the regional mathematical model are revisions to a locked specification, not continuation of the initial model-selection research.

The local 3D URANS–VOF formulation is also established at research-specification level for hydrodynamic barrier-performance assessment. Source roles: ‘3D-RANSVOF’, ‘3D-VOF’, ‘3D-SST’, ‘3D-WAVEBC’, ‘3D-COUPPLING’ and ‘3D-DEFERRED’. Supporting sources are assigned in the relevant Deliverables rather than repeated here as a generic literature list.

10.18 Selected Approach and Rationale

The selected regional mathematical approach is:

horizontal two-dimensional, depth-averaged NLSWE with conserved state $[h, q_x, q_y]^T$, positive-upward bathymetry, bed-slope and Manning sources, dynamic moving bed, open/radiation boundaries, sponge damping where required, and wetting–drying handled by the numerical method

Dispersive extensions, regional-to-local interface algorithms and FSI remain outside this regional mathematical-model lock.

The selected local mathematical approach is:

three-dimensional incompressible immiscible two-phase Navier–Stokes equations, VOF water-volume fraction, mixture density and viscosity, URANS $k-\omega$ SST effective viscosity, regional-to-local inlet forcing, open/radiation artificial boundaries, fixed no-slip terrain and barrier surfaces, and hydrodynamic extraction of pressure, shear, force, moment, run-up, overtopping, transmission and energy metrics

Porosity, trapped-air compressibility, scour, sediment transport, deformable barriers, structural failure and final certification remain deferred or external to the baseline hydrodynamic model.

10.19 Unresolved Decisions

Established	Deferred or separate
Regional 2D NLSWE formulation; conserved and primitive states; bed convention; source-term catalogue; dynamic moving bed; open, reflective and sponge boundary roles	QGIS/geospatial domain lock; exact bathymetry/topography dataset; exact fault-source dataset; validation execution; C++ implementation; local coupling operators; FSI model; optimisation and ML workflow

Table 6. Regional mathematical-model established/deferred status

Established	Deferred or separate
Local 3D incompressible two-phase Navier–Stokes model; VOF phase fraction; mixture properties; URANS $k-\omega$ SST baseline; local boundary partition; one-way replay role; pressure, shear, force, moment, run-up, overtopping and energy outputs	Exact VOF advection algorithm; pressure-correction sequence; numerical wall functions; mesh refinement; validation execution; simultaneous-coupling activation tolerance; FSI, structural failure, scour, sediment, detailed aeration and certification

Table 7. Local three-dimensional mathematical-model established/deferred status

10.20 Requirements and Handoffs

‘RES-NUM-SPEC’ shall carry the corresponding discrete finite-volume specification. ‘RES-DAT-GEO’ and ‘RES-DAT-SRC’ shall close the domain and dataset selections. ‘RES-VER-SPEC’ shall define the lake-at-rest, wet/dry, moving-bottom, boundary and Tōhoku validation tests. Software work shall not treat this document as evidence that implementation or validation is complete.

For the local model, ‘RES-NUM-SPEC’ shall carry the URANS–VOF finite-volume solving sequence and convergence checks. ‘RES-VER-FRM’, ‘RES-VER-BMK’ and ‘RES-VER-MET’ shall validate solver capability, event-conditioned hydrodynamics and local impact/loading separately.

10.21 Deliverable Completion Record

Regional and local hydrodynamic model status: draft complete for research-specification purposes; review, validation and implementation remain pending.

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